Project 5: Data Sorting and Searching Application

In this project, we will create a **Data Sorting and Searching Application** in Python. This application will allow users to sort and search through a set of data using various algorithms, helping them to better understand how these algorithms work and how they can be used to handle different types of data.

Project Overview

The main features of this project will include:

- **Sorting**: Implementing different sorting algorithms to order data in ascending or descending order.
- **Searching**: Implementing various searching algorithms to find a specific element in a dataset.
- **Interactive User Interface**: Providing an easy-to-use interface to input data and choose algorithms.
- **Displaying Results**: Showing the sorted data and the results of the search operations.

We will cover the following sorting algorithms:

1. **Bubble Sort**
2. **Selection Sort**
3. **Insertion Sort**
4. **Merge Sort**
5. **Quick Sort**

For searching, we will implement: